No. of Functional Features Included in the Solution

Date	05 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Functional Features Overview:

This document presents the functional features implemented in the **A Comprehensive Measure of Well-Being** project. The system uses a **Machine Learning (Random Forest Classifier)** model to predict the **Human Development Index (HDI) Category** based on four important development indicators: **Life Expectancy at Birth, Expected Years of Schooling, Mean Years of Schooling, and Gross National Income (GNI) per Capita**. The prediction is provided through a professional **Flask web application** with a responsive user interface..

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Home Page	Displays project introduction and navigation	index.html	Done	High
2	HDI Prediction Form	Accepts user inputs for prediction	index.html	Done	High
3	Machine Learning Prediction	Predicts HDI score using trained model	app.py + HDI.pkl	Done	High
4	HDI Classification	Categorizes result as Low, Medium, High or Very High	app.py	Done	Medium
5	Prediction Result Display	Displays the predicted HDI category clearly on the webpage	result.html	Done	Medium

6	Flask Routing	Connects all webpages using Flask routes	app.py	Done	Medium
7	User Input Validation	Ensures all required fields contain valid numerical values before prediction	HTML Forms	Done	Medium
8	Professional Responsive Interface	Responsive UI with modern design, animated background and CSS styling	HTML + CSS	Done	High

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5
Additional / Nice-to-Have Features	3
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Home Page, Prediction Form, Result Page
2	Backend / Logic	3	Flask Routing, Random Forest Prediction, HDI Classification

3	Machine Learning Model	1	Trained Random Forest Model (model.pkl)
4	API / Integration	1	Flask Request Handling
5	Security / Validation	1	User Input Validation